

Fuel Cell Technologies for Continuous Load Applications

A Position Paper

Version 0.1 — DRAFT

May 2026

Tim Roxey

Eclectic Technologies

Companion to: Powering the Always-On Economy (v0.4) and Generation Mix Optimization Under Competing Objectives (v0.3)

Abstract

Fuel cells are not a primary generation source. They are an electrochemical conversion technology whose performance, carbon profile, and resilience value depend entirely on the fuel that feeds them. For continuous load applications — facilities drawing large, uninterrupted electrical power year-round, of which hyperscale data centers and large-scale desalination plants are the most prominent current examples — fuel cells occupy a specific and valuable architectural position: a second-tier resilience layer beneath primary grid-connected generation, providing onsite ride-through capability measured in days rather than hours. This paper defines the continuous load requirement, positions fuel cells correctly within the generation architecture that the requirement demands, examines the two primary fuel pathways and their implications, and identifies the engineering challenges and policy gaps that currently limit deployment. The conclusion is that hydrogen fuel cells backed by green hydrogen produced from nuclear electrolysis represent the most coherent resilience solution for continuous load applications, complementing rather than competing with nuclear as the primary generation source.

1. The Continuous Load Requirement

Most electricity customers have a load profile — a pattern of consumption that varies with the time of day, the day of the week, and the season. A residential neighborhood draws more power in the evening than at three in the morning. A factory may run a day shift and idle overnight. A university campus is busy during the academic year and quiet in summer. These variations matter enormously to grid planning because they determine how much generation capacity must be available at peak, how much is needed at the baseload floor, and how the gap between the two is managed.

Continuous load facilities are different in kind. They draw the same power at two in the morning as at two in the afternoon, on Monday as on Saturday, in January as in July. The load is not merely large — it is flat. The grid sees no daily cycle, no weekly rhythm, no seasonal swing. The generation serving these facilities must therefore be continuously available, continuously dispatched, and continuously reliable. Intermittency — the characteristic of a source that is sometimes running and sometimes not — is not a cost or efficiency problem for continuous load service. It is a categorical disqualification.

Two facility classes define the current continuous load landscape at a significant scale.

1.1 Hyperscale Data Centers

A hyperscale data center is a facility, typically several hundred thousand square feet in floor area, housing tens to hundreds of thousands of servers operating at continuous full utilization. The electrical demand of such a facility ranges from 100 megawatts to more than 500 megawatts,

drawn continuously. Cooling, power conversion, networking, and compute infrastructure all operate around the clock. The facility has no off-peak mode. Its operators have contractual obligations to the tenants — cloud providers, artificial intelligence infrastructure firms, financial institutions — that make any unplanned interruption a breach event. Many of those tenants have additionally committed publicly to net-zero electricity consumption, making the carbon content of the power supply a contractual as well as an environmental question.

NERC projects data center load growth as the dominant driver of new demand across multiple North American interconnections through at least 2030. The investment community has recognized the opportunity. The constraint that is now limiting development is not land, fiber, or cooling technology. It is the availability of electricity that is simultaneously continuous, affordable, low-carbon, and resilient.

1.2 Large-Scale Desalination Plants

A large seawater reverse osmosis desalination plant is an equally compelling continuous load example, with an additional dimension that data centers do not carry: the cascade consequence of a power interruption is not lost compute but lost water supply.

Modern large-scale SWRO facilities — plants producing hundreds of millions of gallons of potable water per day — draw 100 to 400 megawatts of electrical power continuously. The energy consumption is intrinsic to the process: high-pressure pumps force seawater through semi-permeable membranes at pressures of 60 to 80 bar, and the energy to maintain that pressure is not optional. Advanced energy recovery devices capture and recycle a significant fraction of the pressure energy from the reject brine stream. Still, the net consumption remains 3 to 4 kilowatt-hours per cubic meter of product water under the best current technology, for a plant producing 100 million gallons per day, which translates to a continuous electrical draw of approximately 130 to 180 megawatts with no meaningful load flexibility.

The world's largest desalination installations — in Saudi Arabia, the United Arab Emirates, Israel, and Australia — already operate at this scale. Expanding populations in water-stressed regions, declining aquifer levels, and increasing drought frequency driven by climate change are producing policy commitments to large new desalination capacity across the Middle East, North Africa, southern Europe, the American Southwest, and coastal Australia. Each of these facilities will require a continuous electrical supply of the same character as a hyperscale data center, but with a water security consequence chain that makes reliability failure a public health event rather than a commercial disruption.

The combination of these two facility classes — data centers and desalination plants — establishes a growing and geographically dispersed class of load that existing generation investment frameworks were not designed to serve correctly. The analytical foundation for what continuous load generation requires is developed in detail in the companion documents referenced above. This paper addresses the specific role fuel cells can play within the architecture that requires demands.

2. What Fuel Cells Are and Are Not

A fuel cell is an electrochemical device that converts the chemical energy of a fuel — most commonly hydrogen, but also natural gas, methanol, or other hydrogen-bearing compounds — directly into electricity through an oxidation reaction at the anode and a reduction reaction at the cathode, without combustion. The electrochemical pathway avoids the thermodynamic limits that constrain heat engines, and modern fuel cell systems achieve electrical efficiencies of 50 to 65 percent, with combined heat and power configurations reaching 80 to 90 percent if the thermal output can be used. Emissions from the fuel cell stack itself, when the fuel is pure hydrogen, are limited to water vapor.

These properties make fuel cells attractive. But a fuel cell is not a primary energy source. It is a conversion device. The energy it delivers was stored in the fuel at some earlier point, and that fuel had to be produced, transported, compressed, and stored before the fuel cell could convert it. Every property that matters for the continuous load application — carbon intensity, resilience, cost, and scalability — depends at least as much on the upstream fuel supply chain as on the fuel cell itself.

Evaluating a fuel cell without specifying the fuel pathway is like evaluating a generator without specifying the fuel. The conversion device is not the analysis. The fuel pathway is.

A second property of fuel cells that matters for the grid stability analysis is that they connect to the electrical grid through power electronic inverters. They do not incorporate rotating machinery and therefore contribute no synchronous inertia to the grid — the stored rotational energy that synchronous generators automatically inject into the interconnection when frequency declines. As the generation mix shifts toward inverter-based resources and total grid inertia falls, the stability consequences of large generator trips become more severe. Fuel cells, like solar and wind, do not help with this problem. For the continuous load application, this means fuel cells are correctly positioned as an onsite resilience resource rather than as a grid-stabilizing generation source.

3. The Hydrogen Fuel Pathway Question

Hydrogen is not a primary energy source in the way that uranium, natural gas, or sunlight is. It does not occur in useful concentrations in nature. Every kilogram of hydrogen in commercial use had to be produced from something else, using energy. The carbon intensity of the hydrogen supply chain is the dominant variable in the fuel cell's environmental performance, and the

method of production determines whether a hydrogen fuel cell system is genuinely low-carbon or merely low-emission at the point of use.

3.1 Grey Hydrogen: Current Commercial Standard

Approximately 95 percent of the hydrogen currently produced globally is grey hydrogen — produced by steam methane reforming of natural gas. The reaction converts methane and steam into hydrogen and carbon dioxide. The carbon dioxide is typically vented to the atmosphere. The lifecycle carbon intensity of grey hydrogen is approximately 9 to 12 kilograms of carbon dioxide equivalent per kilogram of hydrogen, which translates to approximately 90 to 120 grams of carbon dioxide equivalent per kilowatt-hour of electrical output from a fuel cell operating on that hydrogen. This is substantially better than burning natural gas directly (approximately 490 grams per kilowatt-hour), but it is not compatible with net-zero electricity commitments. Grey hydrogen in a fuel cell produces clean exhaust; it does not produce clean electricity on a lifecycle basis.

Blue hydrogen addresses the carbon problem of grey hydrogen by capturing the carbon dioxide from the reforming process — carbon capture and storage applied upstream of the fuel cell. If the capture rate is high and the geological storage is permanent, blue hydrogen can achieve lifecycle carbon intensities of 1 to 3 kilograms of carbon dioxide per kilogram of hydrogen, which at fuel cell conversion efficiency translates to approximately 10 to 30 grams per kilowatt-hour — competitive with nuclear and wind on a lifecycle basis. The constraint is that high-capture-rate CCS at commercial hydrogen production scale is still being demonstrated, and the premium over grey hydrogen is substantial.

3.2 Green Hydrogen: The Low-Carbon Pathway

Green hydrogen is produced by electrolysis — passing an electrical current through water to split it into hydrogen and oxygen. If the electrical current comes from a low-carbon source, the resulting hydrogen carries very low lifecycle emissions. Electrolysis using nuclear power produces hydrogen with lifecycle carbon intensity below 1 kilogram of carbon dioxide per kilogram of hydrogen — essentially decarbonized on a lifecycle basis. Electrolysis using wind or solar power produces similarly low-carbon hydrogen, subject to the capacity factor constraints that affect those generation sources.

The current constraint on green hydrogen is cost. Electrolysis at a commercial scale requires capital investment in electrolyzers, compression equipment, and storage, and the electrical energy input represents an ongoing operating cost. At current electrolyzer costs and electricity prices, green hydrogen is produced at \$4 to \$8 per kilogram, compared to grey hydrogen at \$1 to \$2 per kilogram. The levelized cost of electricity from a fuel cell operating on green hydrogen is therefore materially higher than from one operating on grey hydrogen, and materially higher than nuclear generation on a direct comparison basis.

The cost curve is moving. Electrolyzer manufacturing is scaling, efficiency is improving, and the policy environment — the hydrogen production tax credit under the Inflation Reduction Act, European hydrogen strategy investments, and Gulf state hydrogen export programs — is creating demand that is pulling the cost down. The current consensus projection among equipment manufacturers and independent analysts is that green hydrogen will reach \$1.50 to \$2.50 per kilogram by 2030 under optimistic but not implausible assumptions about deployment scale and electrolyzer cost reduction. At that price point, the fuel cell resilience case becomes economically viable in high-value continuous load applications.

3.3 The Nuclear Electrolysis Pairing

The most analytically coherent configuration for the continuous load application pairs nuclear primary generation with green hydrogen production from nuclear electrolysis. Nuclear plants operating in a grid that includes significant variable renewable capacity will periodically face conditions where their output exceeds grid demand — particularly as renewable penetration increases. Rather than curtailing that output or operating at reduced power, the nuclear plant can direct surplus electricity to onsite or co-located electrolyzers, producing hydrogen that is stored for later use in fuel cells. The hydrogen storage then functions as a long-duration energy storage system charged by the nuclear plant and discharged through fuel cells when grid conditions require or when the continuous load facility needs onsite backup.

Nuclear electrolysis closes the loop. The primary generation source charges the resilience layer. The resilience layer provides the ride-through capability that the primary generation source's grid connection cannot guarantee. Together they satisfy all requirements — continuous delivery, net-zero carbon, resilience against supply chain disruption — that neither satisfies independently at the scale continuous load applications require.

This configuration also addresses the capital recovery argument that applies to nuclear generation. A nuclear plant operating above baseload for electrolysis during off-peak periods maintains its revenue stream and productive utilization even when grid demand does not absorb all of its output. The hydrogen it produces is a stored commodity with a market value — either for use in the co-located fuel cell system or for sale into the developing hydrogen market. This diversifies the nuclear plant's revenue profile in a way that pure electricity sales do not.

4. The Two-Tier Resilience Architecture

The continuous load problem requires a generation architecture with two distinct layers, each addressing requirements that the other cannot satisfy alone.

4.1 The First Tier: Primary Grid-Connected Generation

The first tier is the primary generation source — the plant or plants that deliver continuous, grid-connected, low-carbon electricity as the facility’s normal operating supply. The requirements for this tier, as developed in the companion documents, are: availability factor above 90 percent, lifecycle carbon intensity below 20 grams per kilowatt-hour, firm cost of continuous delivery at or below approximately \$105 per megawatt-hour, fuel independence duration measured in months, and positive synchronous inertia contribution to the grid. Nuclear satisfies all of these. No other currently available new-build option satisfies all of them simultaneously.

4.2 The Second Tier: Onsite Resilience Storage

The second tier is an onsite energy storage and conversion system capable of providing full facility load for an extended period without any external input — no grid connection, no fuel delivery, no operator action beyond initiating the islanding sequence. The requirements for this tier are: discharge duration measured in days rather than hours, carbon-compatible fuel supply, startup reliability under adverse conditions, and no dependence on the same infrastructure that a grid-disrupting event would compromise.

Hydrogen fuel cells satisfy these requirements in ways that battery storage does not at the scale continuous load facilities require. A hyperscale data center drawing 200 megawatts requires 200 megawatt-hours of storage for each hour of ride-through. Four hours of battery backup — the current industry standard for Tier IV data center design — represents 800 megawatt-hours of battery capacity, a quantity that is at the upper bound of what is currently commercially deployable at a single facility. Seventy-two hours of battery backup for the same facility would require 14,400 megawatt-hours of battery capacity — approximately ten times the largest single battery installation currently in commercial operation globally. The energy density of hydrogen storage makes extended duration technically and economically more tractable. A 72-hour hydrogen reserve for a 200-megawatt facility requires approximately 1,200 metric tons of hydrogen stored at 700 bar or as liquid, representing a large but engineering-tractable storage system comparable in scale to the liquid fuel storage at a conventional power plant.

For large desalination plants, an additional consideration applies. The cascade consequence of a power interruption — loss of water supply to a population that depends on that plant as its primary fresh water source — means that the resilience requirement is not defined by commercial contract terms alone. The public health consequence of failure defines it. A 72-hour or longer ride-through capability may be a minimum rather than a premium for this facility class, particularly in arid regions where the population’s survival is directly coupled to plant availability.

4.3 What Each Tier Provides

The first tier provides normal operations: continuous, low-carbon, synchronous electricity delivered through the grid connection, with all the stability characteristics that synchronous generation contributes to the interconnection.

The second tier provides abnormal operations: full-facility islanding capability when the grid connection is lost or compromised, sustained for long enough to outlast the events — weather, physical damage, supply chain disruption — most likely to cause the grid disruption in the first place.

The two tiers do not compete. They address different failure modes operating on different timescales. Designing a continuous load facility without the second tier is analogous to designing a hospital with a reliable utility connection but no emergency generator — rational for normal conditions, unacceptable for the conditions the facility must survive.

5. Current State of Technology and Deployment

The technology components of the hydrogen fuel cell resilience layer are individually mature. Proton exchange membrane electrolyzers and fuel cells are manufactured at a commercial scale. Hydrogen compression and storage at high pressure and in liquid form are established industrial processes. The challenge is integrating these components at the scale continuous load applications require, at a cost that is justified by the resilience value delivered.

5.1 What Has Been Demonstrated

Microsoft's Project Natick and subsequent data center fuel cell programs have demonstrated hydrogen fuel cell backup at megawatt scale, operating as a direct replacement for diesel emergency generators, with demonstrated multi-day ride-through capability in controlled conditions. The technical feasibility at the data center scale is established. The commercial case at full continuous load scale — 200 megawatts for 72 hours or more — has not yet been demonstrated in an operational facility.

Large-scale hydrogen storage is operational in the industrial gas and refining sectors, where hydrogen is stored at scale as a process feedstock. The conversion of this industrial storage expertise to the energy resilience application is a systems integration challenge, not a fundamental technology gap.

High-power fuel cell systems for stationary applications are commercially available from multiple manufacturers, including Bloom Energy, Fuel Cell Energy, and Ballard Power Systems, at unit scales of 250 kilowatts to several megawatts. Integrating multiple units to a 200 megawatt scale is an engineering and project management challenge comparable to other large power plant projects.

5.2 What Is Not Yet Solved

Green hydrogen cost remains the primary economic constraint. At current prices, the fuel cost alone for a 72-hour hydrogen reserve at 200 megawatts represents a capital and operating

expense that is difficult to justify on resilience value alone, without policy support or regulatory mandates that assign explicit value to extended ride-through capability.

Electrolyzer durability and maintenance requirements at high utilization rates are still being characterized in commercial deployments. Published stack lifetimes of 60,000 to 80,000 hours are approaching the threshold needed for 10-year system designs, but real-world data at utility scale is limited.

The regulatory and insurance frameworks for large-scale onsite hydrogen storage at industrial facilities are still developing in most jurisdictions. Permitting timelines and safety requirements for multi-hundred-metric-ton hydrogen storage are not yet standardized, creating project development uncertainty that adds to the effective cost.

6. Policy and Regulatory Gaps

Three regulatory gaps currently limit the deployment of hydrogen fuel cell resilience systems at continuous load facilities, independently of the technology and cost challenges.

6.1 No Minimum Ride-Through Requirement for Critical Continuous Load

No current North American reliability standard establishes a minimum onsite energy reserve duration for facilities classified as critical continuous load. Data center standards — including the Uptime Institute Tier IV classification, which is the de facto industry benchmark — require 96-hour concurrent maintainability for critical systems but specify this in terms of system architecture redundancy, not energy storage duration. A Tier IV data center may satisfy all classification requirements with a four-hour battery reserve and a diesel generator that depends on continuous fuel delivery. The four-hour battery covers generator start-up time; the diesel generator depends on a fuel supply chain that may be disrupted by the same event that disrupted the grid.

A minimum ride-through duration standard for critical continuous load, specified in hours of onsite stored energy at full facility load without external input, would create the regulatory demand signal that the hydrogen storage industry currently lacks. This standard does not require specifying fuel cell technology — it sets the performance requirement and allows the market to identify the most cost-effective means of satisfying it.

6.2 No Lifecycle Carbon Standard for Backup Generation

Backup generation at data centers and industrial facilities is currently exempt from the lifecycle carbon accounting frameworks that apply to primary generation. A data center that sources its primary electricity from a nuclear or renewable power purchase agreement may still operate a fleet of diesel emergency generators that, if they run for extended periods during a grid event,

produce emissions inconsistent with the facility’s net-zero commitment. No current regulatory framework requires that backup generation be compatible with carbon commitments, because backup generation has historically been assumed to run only during short emergency events.

As the frequency and duration of grid disruptions increase — driven by extreme weather, aging infrastructure, and the grid stability consequences of the transition away from synchronous generation — the assumption that backup generation rarely runs is becoming less valid. A carbon standard for backup generation that applies to extended-duration events would accelerate the replacement of diesel backup with hydrogen fuel cell systems in facilities with net-zero commitments.

6.3 No Strategic Reserve Framework for Electrolyzer Manufacturing Capacity

The hydrogen resilience architecture depends on electrolyzer manufacturing capacity scaling to meet demand. The current manufacturing base is concentrated in a small number of countries, with significant Chinese market share in some electrolyzer technology categories. A supply chain disruption in the electrolyzer manufacturing base — analogous to the large power transformer manufacturing concentration that creates the vulnerability identified in the companion document — would constrain the deployment of hydrogen resilience systems in exactly the circumstances when demand for them would be highest.

No current policy framework establishes domestic manufacturing requirements, strategic reserve obligations, or supply chain diversity requirements for electrolyzer manufacturing capacity in North America or Europe. This gap is analogous to the transformer reserve gap: visible in advance, addressable through existing policy mechanisms, and currently unaddressed.

7. The Single Finding

Fuel cells are the right technology for the second tier of the continuous load resilience architecture. They are not the right technology for the first tier. The case for fuel cells in continuous load applications is not a case against nuclear generation — it is a case for a two-tier architecture in which nuclear and hydrogen fuel cells each address the requirements the other cannot satisfy alone.

A nuclear plant provides synchronous, low-carbon, continuously dispatched electricity through the grid connection. A hydrogen fuel cell system backed by green hydrogen from nuclear electrolysis provides the extended onsite ride-through capability that grid events, physical damage, or supply chain disruption require. Together, they constitute the complete continuous load generation

architecture. Separately, each leaves a gap. Independent of each other, neither is sufficient.

For data centers and desalination plants alike, the architecture is the same and the stakes are complementary. The data center that loses power loses the compute infrastructure on which the rest of the digital economy depends. The desalination plant that loses power loses the water supply on which the population of an arid region depends. Both deserve a generation architecture designed for the consequences of failure, not merely for the cost of normal operations.

The policy changes needed to accelerate this architecture are identified in Section 6. None requires new legislation. All require regulatory will. The technology is ready to be deployed at scale as the economics become viable. Establishing the performance standards that create the demand signal is the first step, and it is a step that existing regulatory bodies have the authority to take.

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